Agent Reliability at the External-Effect Boundary: Conditional Evaluation and Recovery Guarantees

SIGMA AI-assisted research synthesis

Abstract

How can an agent evaluation distinguish correct external outcomes from convincing completion reports, and when can recovery preserve those outcomes? This paper develops a bounded answer for workflows involving irreversible tool effects, imperfect evaluators, and ambiguous acknowledgments. Conditional measurement arguments separate actual correctness from evaluator acceptance, account for reference-label error, and distinguish independent task sampling from randomized auditing of a fixed execution roster. Recovery arguments identify an information barrier when committed and uncommitted operations look identical, then specify stronger service contracts involving atomic deduplication, retention, and execution-time cancellation.

Counterexamples show why negative status reads, local checkpoints, and stable keys alone need not suffice. The contribution is an analytical evaluation framework and an executable-test specification, not a measured improvement in agent reliability. Its practical significance is to make the assumptions behind reliability certificates and recovery guarantees explicit, while identifying the external evidence and controlled experiments still needed to justify deployment claims.

1 Introduction and problem statement

The central question is what correctness and recovery claims an evaluation can support when an agent acts through tools whose external effects are not fully visible in its transcript. The bounded workload is a logical operation that should produce exactly one irreversible effect. A successful-looking response, an actual effect, and evidence that the effect occurred are separate objects.

The target quantity must be fixed before interpreting a score. For deployment-oriented sampling, reliability is the probability of satisfying a prespecified outcome and safety rubric under fixed task distributions, system versions, environments, budgets, and observation horizons. For a completed execution roster, the target is instead that roster's correctness rate or paired correctness difference; random auditing does not by itself justify extrapolation to future deployment.

The contribution is a conditional account of measurement and recovery, rather than an assessment of general agent capability. It connects evaluator error to uncertainty in reliability estimates and connects recovery safety to the service's actual effect boundary. The available evidence does not establish benchmark validity, real-service conformance, or a causal improvement from any architecture.

2 Methods and evidence

The analysis synthesized the supplied mathematical arguments, explicit state-machine definitions, counterexample schedules, and substantive critical responses. It separated definitions, conditional deductions, proposed tests, and absent empirical evidence. The supplied index was treated as a bounded selection rather than an exhaustive review of the research record.

Measurement arguments were organized by estimand and source of randomness. Deployment-oriented arguments used independently sampled task clusters while allowing dependence among repetitions and evaluators within a task. Finite-roster arguments conditioned on completed executions and located uncertainty in randomized audit selection; reference-label accuracy and evidence soundness remained separate premises.

Recovery arguments were compared at the transition that creates the actual external effect. The synthesis distinguished request issuance, delivery, volatile execution, durable commitment, acknowledgment, and client completion. It retained counterexamples to weakened contracts and distinguished at-most-once safety, eventual progress, deadline completion, truthful reporting, and post-incident containment.

This synthesis uses a bounded selection of the archived research. Unreviewed records could affect its conclusions; the full omission ledger remains available with the generation record.

3 Correctness, observation, and the evaluation target

Reliability at the external-effect boundary requires an evaluation target that separates what an execution accomplished from what the agent or its tools reported. The target adopted here is satisfaction of an independently specified outcome rubric under a fixed execution protocol, including its safety constraints and deadline. Acknowledgments and automated completion judgments are observations of that outcome, not substitutes for it. This distinction determines both what counts as success and which executions may legitimately contribute to a reliability estimate.

Consider a deliberately narrow workload: an agent is instructed to cause exactly one irreversible external effect, such as applying one debit or appending one shipment order. The unit of work is the logical request, rather than an individual tool invocation; several invocations may attempt to fulfill the same request. Let N denote the number of effects applied for that request over the specified execution horizon. The effect-count rubric assigns $Y = 1$ exactly when $N = 1$, with $N = 0$ classified as omission and $N \geq 2$ as duplication. This predicate isolates effect-count correctness; it does not, by itself, establish payload validity or satisfaction of a broader task specification.

Define acknowledgment-based completion separately: $R = 1$ when the agent receives at least one valid acknowledgment. A request can commit while its acknowledgment is lost, leaving a correct external state without acknowledged completion. Conversely, under an ordinary non-idempotent interface, an acknowledged invocation can follow an earlier committed invocation and leave duplicate effects. Thus neither receipt nor absence of an acknowledgment determines the effect-count outcome, and a fluent assertion of completion supplies no replacement for external-state evidence.

For this workload, the proposed outcome oracle is a read-only inspection of an authoritative committed-event ledger after execution. Its role is distinct from the agent-visible tool response: the evaluator uses external-state evidence to determine Y, while the agent may have only messages and local records. The oracle must neither alter the task nor return stale final-state information if its labels are to implement the stated rubric. Correct ledger adjudication and the distinction between pre-commit failure and post-commit acknowledgment loss remain implementation requirements, not demonstrated properties of a tested harness; the supplied evidence establishes an analytical target rather than measured reliability of an agent or benchmark.

The broader evaluation target fixes a task distribution D, agent, model and prompt versions, tool interfaces, environment-generation policy, and execution budget. For each sampled task, the protocol specifies a batch of r runs and selects an index J uniformly from that batch. Let $Y_J = 1$ precisely when the selected run meets the independently specified end-state requirements, safety constraints, and deadline, and let $Y_J = 0$ otherwise. Reliability under this protocol is $p = \mathbb{E}[Y_J]$: expected success per attempted execution, not success conditional on acceptance or success after unlimited retries. Timeouts and crashes remain attempted executions.

The execution horizon is part of the rubric rather than an incidental evaluation setting. A run must meet the specified deadline, and the budget and timeout policy constrain which recovery actions belong to that execution. Moreover, a uniformly selected run from a batch is not automatically equivalent to a fresh isolated run: other batch executions cannot be omitted if they change its outcome distribution. The resulting binary target therefore concerns the stated protocol and horizon; it does not independently summarize delayed harm, persistent-state corruption, or severity-sensitive loss.

The automated completion oracle supplies another variable, Z_J, representing its binary acceptance decision. Keeping Z_J, acknowledgment receipt R, and rubric satisfaction Y_J distinct prevents three different observations from being collapsed into a single completion rate. Reference adjudication is intended to assess actual outcomes rather than merely reproduce the automated oracle's decision, but its correctness cannot be presumed from agreement alone. The definition of reliability remains actual rubric satisfaction even when the available evaluator is imperfect.

For inference about D, the task and its prescribed run batch form the task cluster. Repeated executions within a cluster may be dependent, so their number is not automatically the number of independent observations of task difficulty. The sampling construction considered here assumes independently sampled task clusters and independent cluster-level execution randomness, while allowing dependence among runs of the same task. These assumptions define the scope of subsequent population claims; repeated runs can characterize within-task variability without becoming additional independent task samples.

A finite execution roster defines a different target: the outcomes of the particular recorded executions being assessed, rather than expected performance on newly sampled tasks. Task and run identities must therefore remain explicit, including when the roster contains paired executions of alternative systems. Conditional randomized-audit claims concern selection from that frozen roster and must not be interpreted as establishing independent execution outcomes or representativeness of future workloads. Here this distinction specifies the target of a possible certificate, without asserting an unconditional finite-roster guarantee.

These definitions change the central reliability question from whether an agent appears to finish to whether an attempted execution satisfies its external-outcome rubric within the stated horizon. They also distinguish two possible objects of assurance: expected correctness under a fixed task-sampling protocol and correctness within a fixed execution roster. A higher acknowledgment or acceptance rate is consequently not yet evidence of improved reliability; the improvement must concern the chosen actual-outcome target.

4 Conditional measurement and architectural comparison

Reliability measurement and architectural comparison require different corrections for the gap between evaluator acceptance and actual correctness. Fix a workload distribution D, execution protocol, resource budget, observation horizon, and binary outcome rubric; for a sampled task, schedule $r \geq 1$ runs and select one uniformly to define the target execution. Let $Y \in \{0,1\}$ indicate actual satisfaction of the rubric, $O \in \{0,1\}$ automated acceptance, and $R \in \{0,1\}$ reference certification. The reliability estimand is $p = \Pr(Y = 1)$, not either evaluator's acceptance rate, and no independence among these three variables is assumed.

Write $q = \Pr(O = 1)$, $a = \Pr(O = 1, R = 0)$, and $c = \Pr(O = 1, R = 1, Y = 0)$. The first observable correction removes automated acceptances that the reference does not certify; the remaining correction concerns failures accepted by both evaluators. If a prespecified allowance satisfies $c \leq \eta$, then $\Pr(O = 1, Y = 0) \leq a + \eta$, giving $p \geq \max\{0, q - a - \eta\}$. Here η bounds an unconditional probability over the target execution population, not the failure rate conditional on reference certification; false reference rejections can make the bound conservative without invalidating it.

This lower bound is sharp when the allowance on jointly accepted failures is the only restriction connecting judgments to truth. Holding the observed distribution of (O, R) fixed, assign failure to every execution outside joint acceptance and assign failure mass $\min\{q - a, \eta\}$ within joint acceptance; the resulting reliability equals the lower endpoint. Universal success is also compatible with the same evaluator outputs, so the identified interval is $[\max\{0, q - a - \eta\}, 1]$. Consequently, a tighter conclusion requires additional outcome evidence rather than a different manipulation of the same acceptance frequencies.

Evaluator agreement does not supply that missing evidence. If $O = R = 1$ on every execution, the observed judgments are identical in a world where $Y = 1$ always and a world where $Y = 0$ always; their reliabilities nevertheless differ maximally. Additional judges that reproduce the same errors do not resolve this identification problem. No supplied outcome evidence establishes a useful allowance for an actual reference system, so η must presently be reported as a sensitivity parameter; without a defensible restriction on c, the universally valid choice $\eta = 1$ yields no positive reliability certificate.

A finite-sample version must preserve the task as the independent sampling unit. For N independent, identically distributed task clusters under the fixed protocol, define $X_i = r^{-1} \sum_{j=1}^{r} O_{ij}$ and $\widehat{q} = N^{-1} \sum_{i=1}^{N} X_i$; arbitrary dependence among a task's repeated runs is allowed. To estimate a, separately sample m task clusters from the same distribution, execute the same scheduled-run protocol, and select one run uniformly from each cluster, producing $\widehat{a} = m^{-1} \sum_{k=1}^{m} \mathbf{1}\{O_k = 1, R_k = 0\}$.

This audit selection is not conditioned on automated acceptance, and the construction does not treat the Nr executions as Nr independent observations.

Because the cluster averages lie in $[0,1]$, bounded-variable concentration applies without a run-independence assumption. For $0 < \alpha < 1$, set $t_N = \sqrt{\log(2/\alpha)/(2N)}$ and $t_m = \sqrt{\log(2/\alpha)/(2m)}$; a union bound gives simultaneous inequalities $q \geq \widehat{q} - t_N$ and $a \leq \widehat{a} + t_m$ with probability at least $1 - \alpha$. Thus $L_p = \max\{0, \widehat{q} - t_N - \widehat{a} - t_m - \eta\}$ satisfies $\Pr(p \geq L_p) \geq 1 - \alpha$, conditional on the population allowance and sampling premises. Independence between the two estimation errors is not required for the union-bound step.

Architectural comparison introduces a different estimand and an additional direction of evaluator error. Choose prespecified strata $h = 1, \ldots, H$, nonnegative target weights w_h summing to one, and $n_h \geq 1$ independent task clusters in every positive-weight stratum. Execute candidate system A and baseline system B in isolated copies of the same initial environment, with $r_h \geq 1$ scheduled runs per system and unrestricted dependence within each complete task cluster. For $s \in \{A, B\}$, let $S_{shij} = O_{shij} R_{shij}$ denote joint certification and let p_{sh} be expected cluster-average correctness; then $p_s = \sum_h w_h p_{sh}$ and the target improvement is $\Delta = p_A - p_B$.

This is a task-weighted comparison followed by uniform selection among scheduled runs, not a mixture weighted by the number of executions contributed by each task.

A lower bound on improvement must control candidate false successes and baseline missed successes. For a uniformly selected scheduled run in stratum h, assume $\Pr(S_A = 1, Y_A = 0) \leq \eta_{Ah}$ and $\Pr(S_B = 0, Y_B = 1) \leq \xi_{Bh}$, both unconditional probabilities. The binary identity $Y = S - \mathbf{1}\{S = 1, Y = 0\} + \mathbf{1}\{S = 0, Y = 1\}$ gives $\mathbb{E}[Y_A] \geq \mathbb{E}[S_A] - \eta_{Ah}$ but requires the opposite-sided bound $\mathbb{E}[Y_B] \leq \mathbb{E}[S_B] + \xi_{Bh}$ for the baseline. Therefore, with $d = \sum_h w_h \mathbb{E}[S_A - S_B]$, the comparison satisfies $\Delta \geq d - \sum_h w_h(\eta_{Ah} + \xi_{Bh})$ without assuming independent evaluator errors.

Subtracting two individual reliability lower bounds cannot replace this directional correction. Consider $S_A = 1$ and $S_B = 0$ on every execution while $Y_A = Y_B = 1$: certification-based reliability lower bounds can differ by one although actual improvement is zero. Both systems have no false certification in this construction, but the baseline missed-success probability is one. The counterexample refutes lower-bound subtraction as an improvement argument; it does not describe a measured failure rate or contradict a comparison that includes the baseline allowance.

For sampling uncertainty, define each paired cluster contrast as $X_{hi} = r_h^{-1} \sum_{j=1}^{r_h} (S_{Ahij} - S_{Bhij})$ and its weighted average as $D = \sum_h (w_h/n_h) \sum_i X_{hi}$. Each contrast lies in $[-1, 1]$, and independence across complete clusters gives $\Pr(D - d \geq t) \leq \exp[-t^2/(2V)]$, where $V = \sum_h w_h^2/n_h$. The resulting one-sided radius is $t_\alpha = \sqrt{2V \log(1/\alpha)}$. Repetition counts do not enter this worst-case radius because perfectly correlated repetitions cannot create additional independent task clusters.

Missing certification labels can be handled without discarding their tasks or assuming that missingness is random. Define complete certification variables through the fixed judging protocol, replace every missing candidate certification by zero and every missing baseline certification by one, and retain the scheduled denominators. The resulting observed contrast D_{obs} satisfies $D_{\text{obs}} \leq D$ pointwise, even under outcome-dependent missingness, so the complete-data concentration event remains sufficient. This conservative construction does not make missing truth labels known, justify complete-case deletion, or classify a timeout as an actual failure unless the prespecified rubric does

so.

Combining the directional correction with cluster-level concentration yields

$$L_\Delta = \max\{-1, D_{\text{obs}} - \sum_h w_h(\eta_{Ah} + \xi_{Bh}) - \sqrt{2\log(1/\alpha)\sum_h w_h^2/n_h}\}$$

and conditional coverage $\Pr(\Delta \geq L_\Delta) \geq 1 - \alpha$. The candidate and baseline allowances have not been calibrated against actual outcomes in the supplied evidence; agreement among judges does not establish either premise. If a separate valid procedure instead supplies simultaneous upper confidence bounds on the required allowances with failure probability at most β, combining the calibration and evaluation events gives coverage at least $1 - \alpha - \beta$, without requiring independence between those procedures.

The same error allocation applies when the individual reliability allowance is estimated rather than fixed.

These population statements depend on distribution-matched task sampling and independence across complete clusters, neither of which is established for a real benchmark or deployment by the supplied evidence. Shared services or persistent state can undermine those premises, and the bounds do not automatically transfer across task distributions, environments, versions, or budgets. The paired result also concerns a fixed comparison: multiple endpoints, adaptive monitoring, or unrestricted search across mechanisms require separately justified error control. These restrictions distinguish the conditional probability statement from a general claim of architectural effectiveness.

Random auditing of a frozen execution roster answers a separate question and does not require independent execution outcomes. After a fixed roster of M paired executions has finished, the target is its actual success-rate difference, $\Delta_M = M^{-1}\sum_{i=1}^{M}(Y_{Ai} - Y_{Bi})$, rather than an expectation over newly sampled tasks. A uniformly random audit of m pairs supplies the inferential randomness conditional on that completed roster. Dependence among the executions is therefore compatible with a finite-roster certificate, but the certificate alone does not generalize to future tasks.

For this finite-roster construction, hypergeometric inversion supplies marginal bounds on the relevant outcome counts, and a union bound provides simultaneous coverage for the contrast calculation. Unresolved audit labels can be retained through conservative possible-outcome sets rather than resolved by assertion; when those sets contain the actual outcomes, they can only weaken the reported lower bound. Actual-outcome containment is a substantive premise that has not been demonstrated here, and judge agreement does not establish it. The roster audit thus replaces an execution-independence assumption with a randomized-selection argument, not with an assumption-free correctness oracle.

The measurement consequence is that a reliability certificate and an improvement certificate require different evidence: the former controls jointly accepted failures, whereas the latter must additionally constrain how much baseline correctness exceeds baseline certification. A positive corrected lower bound $L_\Delta > 0$ would support improvement for the specified population and protocol under its premises, while a positive finite-roster bound would concern only the completed roster. No empirical audit or controlled architectural comparison is supplied here; the analytical contribution is to identify which outcome evidence and sampling conditions could make an observed acceptance advantage informative about correct external outcomes.

Table 1: Distinct evidence contracts for individual reliability, paired population improvement, and a frozen-roster comparison.

Inference target	**Source of inferential randomness**	**Required connection to actual outcomes**	**Scope of the certificate**
Individual reliability p	Independent, distribution-matched task clusters; repetitions within a task may be dependent	An allowance on jointly accepted failures, $\Pr(O = 1, R = 1, Y = 0) \leq \eta$	Correctness under the fixed workload, protocol, budget, and horizon
Paired population improvement Δ	Independent complete task clusters in prespecified weighted strata; within-pair and within-task dependence allowed	Candidate false-certification allowances and baseline missed-success allowances	The fixed candidate–baseline comparison for the specified task population
Frozen-roster improvement Δ_M	Uniform random audit selection after the paired roster is fixed; execution independence unnecessary	Sound audit outcome sets containing actual outcomes, including unresolved cases	The completed roster only; no automatic population or deployment generalization

Evidence: [Research session 4, artifact 1], [Research session 5, artifact 3], [Research session 6, artifact 1]; claims: C2, C3, C4, C5, C23

5 Information limits after ambiguous commitment

Ambiguous commitment creates an information limit: recovery decisions based on identical local observations can require incompatible actions to preserve the intended external outcome. The workload considered here is one logical operation intended to produce exactly one irreversible external effect, with no deduplication between ordinary requests. The unit of analysis is that operation, and the outcome oracle is its actual effect count, N, with correctness defined as $N = 1$ rather than receipt of a plausible completion message. The first argument evaluates the count after a terminal recovery choice and any stipulated retry commitment; the later delayed-execution arguments also consider effects occurring after the recovery decision.

An independent evaluator would therefore need authoritative external-effect evidence that the recovery policy itself does not possess.

Consider two histories ending at the same timeout. In H_0, the original request is lost before

commitment, leaving $N = 0$; in H_1, it commits, leaving $N = 1$, but its acknowledgment is lost. The accessible checkpoint, messages, and other observations are identical, and no authoritative lookup or distinguishing observation is available before the decision. A policy restricted to this information must consequently assign the same retry probability, q, in both histories, provided its randomization distribution does not depend on the hidden external state.

The forced-choice premise makes the conflict precise: the policy must either stop without retrying or issue one retry that commits exactly one additional effect. In H_0, stopping omits the required effect, whereas retrying produces the correct count; in H_1, stopping preserves the correct count, whereas retrying duplicates the effect. The corresponding error probabilities are $e_0 = 1 - q$ and $e_1 = q$, so $e_0 + e_1 = 1$ and $\max(e_0, e_1) \geq 1/2$. Choosing $q = 1/2$ attains equality for external-effect correctness under these premises, but does not establish equally strong guarantees for independently confirmed completion or deadline satisfaction.

This worst-case statement does not assume that the two histories are equally frequent. If the conditional probability of H_1 among ambiguous timeouts is p, equal weighting of omissions and duplications gives expected error $(1-p)(1-q)+pq$, whose minimum over retry policies is $\min(p, 1-p)$. A policy can thus perform well on this conditional average while failing certainly in the rarer history. Neither the history frequencies nor an appropriate severity-sensitive loss function is supplied by the construction, so the bound is not a deployment failure-rate estimate or a practical recommendation to randomize retries.

Guaranteed retry commitment is essential to the displayed probability bound. If an issued retry can disappear without committing, success in H_1 no longer requires refraining from retry: a dropped retry leaves the already correct effect count unchanged. For example, a policy that always retries has $q = 1$, yet if every retry is dropped, H_1 retains $N = 1$, contradicting the proposed extension $\Pr(\text{success} \mid H_1) \leq 1 - q$. That delivery-relaxed generalization is withdrawn; the original forced-choice argument remains conditional on a chosen retry actually committing, and no replacement numerical bound follows here under weaker delivery assumptions.

The limitation concerns unavailable external information, not the sophistication of local reasoning. Restoring a checkpoint does not reverse an irreversible effect, and replanning from the same accessible information does not distinguish the two histories. Authoritative lookup, escalation, compensation, abstention, and indefinite waiting change the available actions or observations and therefore lie outside the forced-choice result. The histories and bounds are analytical constructions rather than observed incidents or benchmark results; matching implementation transcripts, an independent effect oracle, actual service compliance, and controlled recovery benefits have not been demonstrated.

Even an accurate status interface need not supply the information required for safe replacement. In the extended model, the original request r may remain pending after timeout, each dispatched request executes at most once, and a distinct replacement r' can independently create another effect. Assume that status(r) is linearizable with actual commitment: it returns COMMITTED with a durable result if commitment precedes its linearization point, and otherwise returns NOT_FOUND. There is no stale-read or recording-gap defect in this specification, but NOT_FOUND neither identifies pending work nor prevents its future execution.

The legal ordering

$$\text{status}(r) < \text{commit}(r') < \text{commit}(r)$$

therefore produces two effects after an accurate negative read. The replacement commits first, and the delayed original commits afterward; duplication depends on both commitments, not merely on issuing a retry. Any finite sequence of negative reads before replacement can also occur when the original is permanently discarded, so waiting for more read-only absence observations does not by itself distinguish discard from sufficiently long delay. This counterexample refutes the universal sufficiency of negative commitment reads, but does not transfer the earlier 1/2 bound to the changed execution model.

Wrapper-local cancellation encounters the same distinction between current absence and permanent exclusion. Here, sound CANCELLED authorization at time t means that the original has produced no effect by t and cannot produce one in any admissible continuation; COMMITTED must certify an actual effect, not merely a local dispatch record. Conclusive-resolution liveness requires an eventual known COMMITTED or sound CANCELLED outcome on every admissible execution. The conditional obstruction concerns a downstream service that may discard or arbitrarily delay an already dispatched request and does not consult the wrapper's records, epochs, or cancellation state when producing the effect.

In an execution where the original is permanently discarded, truthful COMMITTED is unavailable, so conclusive resolution requires CANCELLED at some finite decision point. A second execution can reproduce the same wrapper records, clocks, fencing epochs, and absence replies up to that point while delaying the original downstream execution until afterward. The same authorization is then unsound because the original can still commit; if a replacement also executes, the ordering is CANCELLED, replacement effect, then original effect. Durable local records do not exclude this continuation, and the cancellation claim is already false even if no replacement ultimately commits.

The wrapper argument is a conditional safety–progress obstruction, not a universal impossibility of recovery. Waiting or returning UNKNOWN can avoid unsafe replacement without satisfying the specified conclusive-resolution requirement, and fixing policy randomness in the paired histories does not establish an almost-sure impossibility theorem or a numerical error bound. What changes in the central reliability argument is the required content of recovery evidence: observing no past commitment is insufficient when outstanding work can still execute.

Safe replacement requires both a justified account of the original's past outcome and exclusion of its future effect; stronger effect-boundary enforcement can change those premises, whereas local recovery state alone cannot.

6 Atomic recovery contracts and crash semantics

Atomic coupling of an external effect with its completion record supplies a conditional recovery contract: retries can preserve at-most-one effect without ensuring that an effect occurs or that the client learns of it. The workload considered here is a single intended ledger append with stable operation key k and full payload p; the unit of analysis is that logical operation across its deliveries, retries, and crashes. The evaluator inspects the durable effect multiset E directly and checks its

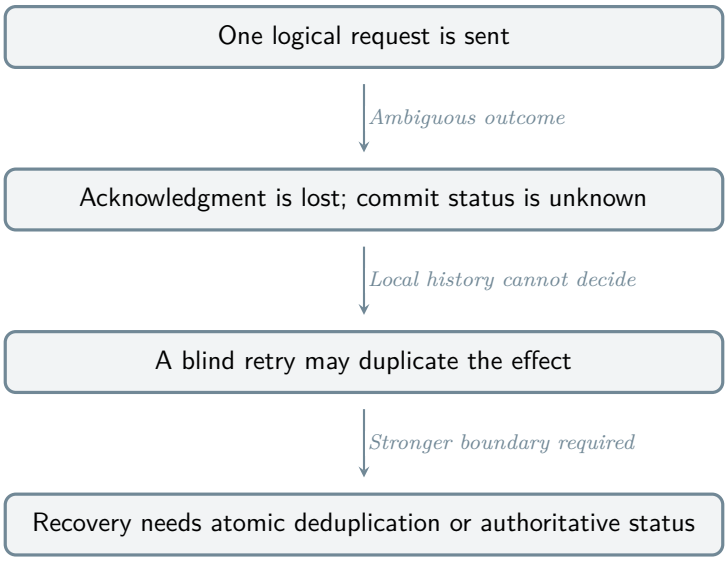


Figure 1: An ambiguous acknowledgment leaves two externally different outcomes with the same local observation. A safe recovery claim therefore depends on a service contract that can prevent or identify a duplicate effect; the diagram is an analytical model, not a verified implementation.

Source-linked claims: C8, C9, C10; evidence links appear in the claim notes.

correspondence with the completion map D, rather than treating a completion record as its own outcome oracle.

Correctness requires exactly one matching effect and no unintended effect at the declared observation horizon, whereas the safety argument concerns every finite execution prefix.

The effect boundary is deliberately narrow: the covered effect is the committed ledger row, not a payment, email, or other downstream consequence. Atomicity, serialization, crash durability, record retention, stable identity, and independent observation are premises of the model, not properties established for an inspected service. The resulting argument and crash traces are analytical specifications, not observed benchmark results or measured reliability improvements; their application to an implementation remains unresolved.

The zero-effect base case is essential. Initially, E and D are empty, so the operation has neither an effect nor a completed record; the argument does not begin from an arbitrary state containing an unrecorded effect. A completed entry $D[k]$ stores the full payload, result, and effect identifier, while retries preserve k and p even though each transport delivery receives a fresh identifier. In the proposed ledger schema, the effect table deliberately permits multiple rows with the same logical key, so duplicate prevention must follow from the handler rather than being assumed by the outcome representation.

Atomic processing has three branches. When $D[k]$ is absent, one durable transaction creates an effect and its matching completion record together; when the record contains the same full payload, processing returns the stored result and identifier without creating another effect. A different payload is rejected without an effect. Serialization or uniqueness conflicts must abort the entire competing transaction, including its effect insertion, and an ambiguous commit outcome must lead

back to a fresh record lookup rather than being treated as a known rollback.

Let N_k count committed effects for key k. The candidate invariant is stronger than merely $N_k \in \{0, 1\}$: either the completion record is absent and $N_k = 0$, or the record identifies exactly one effect with the matching key and payload. Starting from the empty state, the first successful absent-key transition creates the matching pair; replay and rejection preserve it, and a crash exposes either the complete before-state or the complete after-state. Transport transitions do not alter that pair. This inductive argument therefore supports per-key safety under the stated premises, rather than an unconditional guarantee about agent recovery.

Concurrency requires the record test to occur at the serialized processing point, not merely when a request is received. Two workers may both be ready, but the worker processed second must inspect the record created by the first and replay its result. In a transactional realization, a competing absent-key insertion must instead lose serialization or uniqueness arbitration and roll back its effect write. Thus receiving a request does not reserve permission to perform an effect, and an earlier observation of an absent record cannot authorize an independent later append.

Safety does not supply existence: dropping every request leaves $N_k = 0$ while satisfying at-most-once safety. Exactly one effect additionally requires at least one matching request to complete processing, either creating the effect or observing its existing record. Verified client completion requires further conditions on response creation, response delivery, and durable recording of client completion. None of these eventual conditions alone establishes completion by a deadline; the service can preserve the effect invariant while the client remains unable to verify success within its execution budget.

Reproducible crash schedules also require an explicit request lifecycle. Partition the pending-envelope pool into requests Q_R and acknowledgments Q_A, with volatile workers held in V. A request envelope (x, k, p) has a fresh delivery identifier x distinct from the stable operation key; its acknowledgment has a fresh envelope identifier and correlates its outcome with x. The envelope pool survives service crashes but permits explicit permanent drops, while delay leaves an envelope eligible for later processing. This is a destructive-dequeue model, not a broker model with automatic redelivery.

RECEIVE removes an envelope from Q_R and transfers it exclusively into a ready worker in V. PROCESS evaluates the current completion record and converts that worker into a processed worker carrying the outcome; it does not yet create an acknowledgment envelope. ACK_ENQUEUE subsequently creates the acknowledgment in Q_A and removes the processed worker. CRASH clears volatile workers without reconstructing their consumed deliveries, while preserving durable effects, completion records, and envelopes still queued. The table makes the resulting boundaries explicit for one initially unperformed operation.

The effect-first counterexample separates these atomic requirements from a superficially similar two-write handler. The first delivered attempt checks an absent record, commits its effect, and crashes before writing completion; after restart, a second delivered attempt again sees no record and commits another effect. The resulting $N_k = 2$ is a possible safety violation of this particular protocol, even with a stable key and unchanged payload. The counterexample requires the retry actually to be processed: issuing a retry alone does not create the second effect.

Marker-first handling fails differently. A naive handler can durably write a purported completed record containing a reserved effect identifier, crash before performing the effect, and then return cached success on retry despite $N_k = 0$. A more conservative marker-first protocol can instead use a reservation gate: only its uniquely confirmed creator may attempt one append, retries encountering the gate return an unknown or suppressed outcome, and restart never resumes the append. Under those rules, a crash between reservation and append can permanently omit the effect while preserving at-most-once safety; the reservation is not evidence of completion.

The broader assertion that separating the writes necessarily destroys at-most-once safety must therefore be withdrawn. Effect-first duplication and marker-first omission are local counterexamples to different parts of the stronger effect-and-completion contract, not a refutation of every non-atomic recovery protocol. Atomic effect-and-record insertion remains a sufficient conditional design for preserving the covered outcome across retries, while explicit dequeue and acknowledgment semantics determine what recovery must still accomplish. The central argument consequently shifts from treating deduplication as completion to distinguishing preserved safety, actual effect existence, and verified deadline-bounded completion.

Table 2: Predicted crash-boundary states under destructive dequeue and atomic effect–record insertion. Each row considers the original delivery for one initially unperformed operation, with no additional deliveries or explicit envelope drops. These are model-derived outcomes, not executed fault-injection results.

Crash boundary	**Durable effect and completion state**	**Original request after crash**	**Original acknowledgment after crash**
Before RECEIVE	$N_k = 0$; $D[k]$ absent	Remains in Q_R and is eligible for later receipt	Not created
After RECEIVE, before PROCESS	$N_k = 0$; $D[k]$ absent	Consumed delivery is lost with the ready worker; no automatic redelivery	Not created
After PROCESS commits, before ACK_ENQUEUE	$N_k = 1$; $D[k]$ identifies the matching effect	Consumed delivery is lost with the processed worker	Not created; the crash does not reconstruct it
After ACK_ ENQUEUE, before acknowledgment delivery	$N_k = 1$; $D[k]$ identifies the matching effect	Processing and acknowledgment enqueueing are complete; no worker remains	Remains in Q_A, subject to subsequent delivery or explicit drop

Evidence: [Research session 4, artifact 2], [Research session 4, artifact 4]; claims: C10, C12

7 Retention, cancellation, and containment

Retention and cancellation constrain recovery only when they prevent later effect-capable execution, rather than merely stop the agent from sending requests. The unit of analysis here is one logical operation with an immutable payload and stable identity, including its retries, queued copies, and resumed workers; the outcome oracle is its actual irreversible-effect count, denoted $C(t)$, rather than a completion response or the presence of a receipt. At-most-once safety requires $C(t) \leq 1$ at every execution prefix, including after an evaluation horizon H, whereas effect completion by that horizon requires $C(H) = 1$.

The arguments below distinguish conditional safety contracts and constructed counterexamples from implementation evidence: no independently audited execution or real-service validation establishes the stipulated atomicity, timing, or universal execution gating, which would require observation of actual effects independently of receipts and client reports. [C13–C15, C18, C19]

Finite receipt retention invalidates an unrestricted inference from stable identity and atomic effect-plus-receipt commitment to at-most-once execution. In the constructed schedule, the first effect commits at time 1 and its receipt has retention duration $T = 10$, so suppression expires at time 11; an outstanding request bearing the same key executes at time 12 and creates a second effect. The client can already have stopped issuing retries without preventing this duplication, because stopping issuance does not withdraw requests that remain capable of executing. Likewise, an ingress cutoff is insufficient if an accepted request can remain queued until after suppression expires.

The counterexample retracts the unrestricted finite-retention guarantee, not the conditional argument that applies when suppression covers every execution opportunity. Once a receipt expires, its absence is no longer evidence that the operation never committed. Thus, increasing confidence in client shutdown or preserving the original key does not address the decisive gap: the service must still distinguish an unperformed operation from a previously performed operation whose receipt has been removed, or reject execution without needing that distinction.

The first sufficient protection contract retains suppression through every possible effect-capable execution. If a first commitment occurs at time c and its receipt expires at $c + T$, every later transition that could create another effect must occur strictly before $c+T$ and inspect the retained record under the stipulated serialized check-and-commit contract. The strict boundary matters because a transition at expiry cannot rely on a record that is no longer valid. Establishing this premise requires a bound on the last possible processing of every copy, including delayed, redelivered, and resumed requests; an average latency, client send cutoff, or ingress deadline does not establish that bound.

The second protection contract separates result retention from duplicate suppression. A durable tombstone installed atomically with commitment can continue to prohibit execution after the result itself expires; alternatively, an enforceable rejection boundary can eliminate later effect opportunities. The safety requirement concerns persistent suppression or rejection, not indefinite availability of the returned result. Neither protection contract guarantees that an effect occurs, that its result remains retrievable, or that completion is reported by a deadline.

Concurrent workers require a refinement of these protections because eligibility and receipt obser-

vations can become stale before commitment. In the supplied concurrent contract, every effect-producing path uses the same immutable authorization interval $[a, D)$, with $0 < D - a \leq T$, and a first commitment at time c creates a receipt valid on $[c, c + T)$ without renewal on replay. Current execution eligibility, current receipt inspection, and actual effect-plus-receipt commitment must share one per-operation ordering, with garbage collection respecting receipt validity. A worker's earlier observation that execution was eligible or that no receipt existed cannot substitute for these checks at the effect boundary.

The conditional safety argument then partitions every later effect attempt at the authorization deadline. Since the first commitment satisfies $c \geq a$, its suppression interval reaches at least the end of authorization: $c+T \geq a+T \geq D$. Any later attempt before D must therefore encounter retained suppression, while an attempt at or after D must be rejected by the execution-time eligibility check. When expiry equals the deadline, either ordering of garbage collection and an attempted execution is safe only because execution is already ineligible; a paused worker cannot carry a pre-deadline authorization check across that boundary. This strengthens the earlier serialized argument without making finite retention alone sufficient.

Cancellation provides a different certificate: under a stronger contract, it can establish that an initially unperformed operation has neither executed nor retained any future opportunity to execute. Let its durable state S be PENDING, APPLIED, or CANCELLED, with the invariant that $S \in \{\text{PENDING}, \text{CANCELLED}\}$ implies $C = 0$, while $S = \text{APPLIED}$ implies $C = 1$. Cancellation and actual-effect commitment must compete as crash-atomic transitions out of PENDING, and every effect-producing path must obey this state gate. A durable CANCELLED acknowledgment then certifies both the absence of prior commitment and the exclusion of later gated effects.

The acknowledgment's meaning depends on which transition wins, not on which request the client sends first. If effect commitment wins, an ALREADY_APPLIED response reports that outcome; it is not cancellation or rollback. If cancellation wins, queued and resumed workers must remain unable to commit, including after crashes or client recovery, so cancellation suppression must cover every later execution opportunity. Client stopping, a cached eligibility check, an acknowledgment issued before durable suppression, or cancellation state that expires while work remains executable cannot supply this certificate.

Post-incident closure has a deliberately weaker historical claim but can apply when cancellation's clean-state premise no longer holds. A durable CLOSED gate, checked atomically at every covered effect-creation boundary, conditionally prevents additional covered effects after closure commits and until authorized reopening, even when the historical effect count already exceeds one. Unlike cancellation from PENDING, closure need not certify that no prior effect occurred. Its contribution is to stop further growth of the covered effect history, not to restore an at-most-once property that has already been violated.

The containment boundary also limits what closure can repair or prevent. Historical committed effects remain part of the count even if refunds or corrective records are later created, and closure alone establishes neither successful compensation nor notification receipt. Already accepted downstream work remains outside the guarantee unless its eventual effect-creation path also enforces the gate. Consequently, the recovery argument must distinguish three achievements: preserving at-most-once execution through retention or rejection, certifying zero effects through competing

cancellation, and containing additional effects through closure; none can be substituted for repairing historical harm.

Table 3: Conditional guarantees differ in their historical premises and temporal scope. These are analytical contracts, not measured implementation outcomes.

Contract	**Required starting condition**	**Certified effect property**	**What it does not establish**
Retention-protected or durably rejected execution	Initially unperformed operation; all effect paths obey the applicable atomic suppression and execution rules	At most one effect across execution prefixes, with protection extending through every effect-capable opportunity	Existence of an effect, result availability, or deadline completion
Durably acknowledged cancellation	Initially unperformed operation; cancellation and actual commitment compete from PENDING	Zero prior effects and no future gated effects after CANCELLED	Rollback when execution has already won
Durable closure	Historical effects may already exist, including duplication; all covered future effect transitions obey the closure gate	No additional covered effects after closure commits while the gate remains closed	Removal of historical duplication, compensation, notification receipt, or containment of ungated downstream work

Evidence: [Research session 6, artifact 4], [Research session 6, artifact 5], [Research session 6, artifact 7], [Research session 6, artifact 6], [Research session 7, artifact 7]; claims: C14, C15, C18, C19

8 Evidence contracts and truthful reporting

Evidence-based certification requires more than a correctly implemented evaluator: the evidence must describe the covered history truthfully, and the resulting outcome set must contain the actual outcome. This distinction identifies where a reliability certificate can fail even when its mapping rules are followed exactly. It also exposes a separate temporal requirement: evidence of correctness at an execution horizon need not establish that a completion report was true when emitted.

For this analysis, the workload is an append operation in an initially empty, isolated namespace with expected payload p. The unit of adjudication is one scheduled execution slot, whose history w orders durable append commits, accepted final-report receipts, and cutoff H. The external-effect

oracle assigns $E_H(w) = 1$ exactly when one covered append commit occurs before H and contains p; the reporting oracle assigns $R_H(w) = 1$ exactly when one final-report receipt occurs before H and is a well-formed, matching SUCCESS.

The combined outcome is $Y_H = E_H \wedge R_H$; additional commits or final receipts violate the corresponding exact-one requirement, and nonexecution remains in the scheduled roster with outcome $Y_H = 0$ under the specified operational rule. The evaluators considered below are an evidence-to-outcome-set mapper and a reference adjudication used to check its evidence, not the agent's own completion assertion. [C6; [Research session 7, artifact 9]]

The conservative mapper assigns nonempty component sets $U_E, U_R \subseteq \{0, 1\}$ from routine evidence d and authorization state a, then returns $S = m(d, a) = \{e \wedge r : e \in U_E,\ r \in U_R\}$. A usable complete certificate determines a component by applying its full rubric to the certified contents. A sound record of an irreversible violation, such as two distinct covered commits, can establish failure without complete enumeration; by contrast, observing one matching commit cannot establish that it was the only commit. Missing evidence otherwise produces $\{0, 1\}$, while recognized contradictions or quarantined authorities widen every affected component to $\{0, 1\}$ rather than producing an empty set.

Four predicates separate implementation behavior, evidence truth, outcome coverage, and authorization. Mapper conformance, $M(d, a, S)$, means that the frozen output equals the specified mapping for the evidence and authorization version actually supplied, including required conflict and hold flags. Evidence-contract validity, $V(d, w)$, means that the certificates' assertions about contents, completeness, namespace, and cutoff placement are true of the covered history; actual-outcome containment, $Z(S, w)$, means $Y_H(w) \in S$.

A certification hold, $B_t(D)$, instead concerns the information and authorization available at review time t: certification depending on trust domain D is suspended when that domain is unauthorized, a relevant contradiction remains unresolved, or a disclosed breach has not been bounded and independently revalidated. A hold neither proves task failure nor repairs false evidence.

Under the stipulated scope, evidence validity and mapper conformance jointly imply containment. A truthful complete certificate places the actual component value in its assigned set; a sound irreversible-violation witness correctly places failure in that set; every remaining branch admits both values. Consequently, $E_H \in U_E$ and $R_H \in U_R$, so applying conjunction to the actual pair gives $Y_H \in S$. The component product can admit combinations that no history realizes, but this only enlarges the output set and therefore does not undermine containment.

The converse implications do not follow. Conformance constrains the relation between evidence and output, not the truth of the evidence, so a false complete certificate can make a conforming mapper return $\{1\}$ when the actual outcome is 0. Valid evidence likewise does not guarantee conformance if an implementation ignores it or computes the wrong set, and containment can hold accidentally despite an invalid certificate. The analytical fixtures in the table make the first failure mode explicit: an omitted extra commit or final receipt defeats the outcome rubric without creating a mapper implementation defect. These are reconstructed counterexamples, not observed incidents or recovered historical traces.

Disclosure changes the certification decision, not the correctness of the frozen mapping relative to

its original inputs. In the silently false fixtures, evidence validity and containment were already false before disclosure, although the reviewer did not know this; a sound reference reveals those failures without retroactively making mapper conformance false. Remediation therefore preserves the original evidence and output, creates a new evidence version, and quarantines affected authorities. The replacement mapping widens affected components while retaining any independently supported component, but conservative remapping does not restore the original certificate's validity or by itself justify lifting the issuer hold.

This separation supplies the evidence premise needed for randomized auditing of a frozen paired roster. Conditional on sound outcome sets and the specified uniform audit selection, finite-roster coverage can concern audit randomization even when execution outcomes are correlated; independent execution outcomes are not required for that conditional target. Unknown labels can be handled conservatively, but a falsely definite label that excludes truth violates the premise rather than merely reducing precision. Such coverage is neither a posterior probability that the particular roster is correct nor a deployment-reliability claim.

Across this analysis, certificate soundness, observer completeness, operational audit compliance, and implementation conformance remain unvalidated: no executed pilot, controlled agent comparison, or measured reliability improvement is supplied. [C5, C6; [Research session 7, artifact 1]]

Report-time truthfulness requires a further outcome predicate because the horizon rubric permits commitment after reporting. Let X_h denote exactly one correct covered commit by cutoff h, and let Y_{window} additionally require the horizon outcome and correctness at a later cutoff. In contrast, let T require both the full reporting contract and the reported effect's presence as the unique correct covered effect immediately before report emission.

These predicates distinguish eventual correctness within the declared horizon from the truth of the completion assertion at its own event boundary; they do not equate truth with the agent's justified knowledge or with protection against future effects. [C7; [Research session 7, artifact 8]]

An analytical temporal control demonstrates the distinction. Suppose a unique, timely, well-formed matching SUCCESS is emitted at tick 15, the sole correct commit occurs at tick 18, and no other covered commit occurs through tick 40. With horizon $H = 20$ and the stipulated receipt contract satisfied, the outcome vector is

$$(X_{20}, X_{40}, Y_H, Y_{\text{window}}, T) = (1, 1, 1, 1, 0)$$

. Moving the sole commit to tick 14, with the remaining conditions unchanged, makes the vector $(1, 1, 1, 1, 1)$.

The late-commit control therefore passes both horizon and later-state checks while failing report-time truthfulness; it is a counterexample to treating those checks as a truthfulness certificate, not a refutation of a client contract that requires reporting after successful commitment.

A conservative temporal evaluator can express uncertainty by considering the histories consistent with retained evidence. If $W(d)$ is that history set and P an outcome predicate, define $O_P(d) = \{P(w) : w \in W(d)\}$; actual-trace membership then implies containment of the actual predicate value. Joint predicates must be evaluated on the same histories so that their dependence is preserved. A singleton success label for either horizon completion or report-time truth must include the entire

reporting contract—content, uniqueness, timeliness, and complete capture or closure—not merely one matching row and one observed SUCCESS.

A complete capture containing INDETERMINATE, a late SUCCESS, or multiple final reports fails that reporting contract despite a correct external row.

Ordering uncertainty must also be established from what the retained evidence genuinely permits. Removing a separate ordering certificate does not make report-time truth unknown if trusted timestamps on a common clock still establish emission at tick 15 and commitment at tick 18. Conversely, same-tick timestamps without within-tick ordering need not determine whether the effect existed immediately before emission; an uncertainty label is justified only when the admissible histories actually permit both outcomes.

The evidence therefore changes the central reliability argument in two specific ways: a conforming evaluator supports certification only through a sound containment contract, and correct end states support truthful completion claims only when the report-time boundary is adjudicated separately.

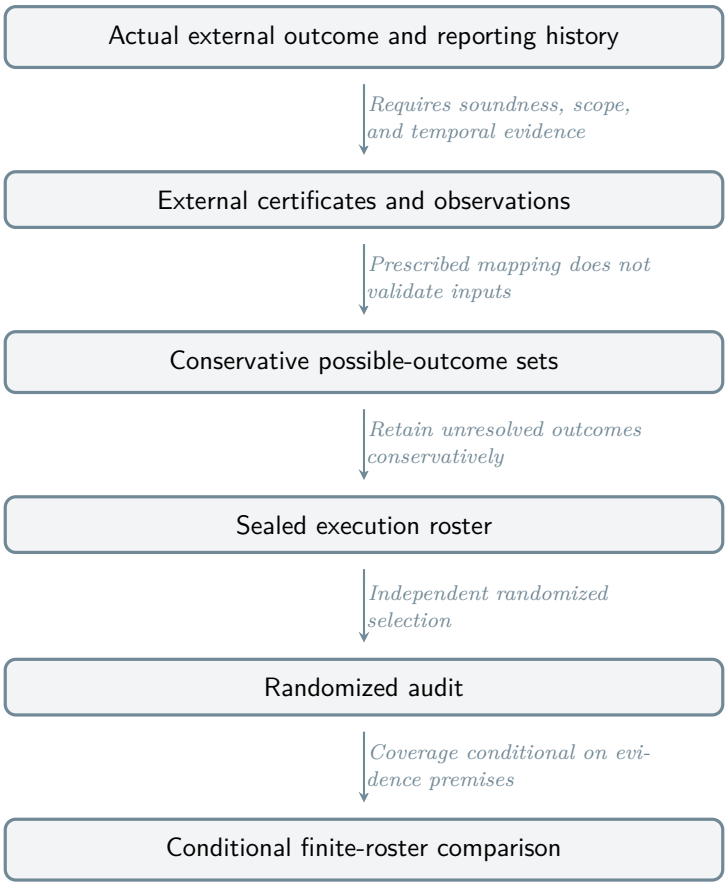


Figure 2: The conditional evidence chain separates external outcomes, evidence validity, mapper conformance, and randomized auditing. None of these dependencies establishes that a practical implementation satisfies the required contract.

Source-linked claims: C5, C6, C7, C20; evidence links appear in the claim notes.

Table 4: Conditional analytical fixtures separating mapper conformance, evidence validity, containment, and certification holds. Each fixture assumes an initially authorized issuer and a conforming mapper. The silently false fixtures are reconstructions, not observed incidents or identified historical cases. Hold decisions are shown before and after sound reference disclosure; the frozen mapping is unchanged.

Fixture	**Stipulated evidence and history**	**Frozen output** S	**Conformance** M	**Validity** V	**Containment** Z — $Y_H \in S$ **implies the set contains truth, not necessarily success** $Y_H = 1$ **in general; here shown directly as true or false — actual** Y_H **noted in the history column where false evidence matters**
Clean evidence	Exactly one correct commit and one valid final receipt precede H; matching complete certificates. No hold before or after disclosure.	$\{1\}$	True	True	True
Visible conflict	Certificates assert incompatible exact counts for the same namespace and cutoff. Hold before and after disclosure.	$\{0, 1\}$	True	False	True
Silent commit omission	Certificates assert a clean history, but omit a second pre-cutoff commit; actual $Y_H = 0$. No hold before disclosure; hold afterward.	$\{1\}$	True	False	False
Silent receipt omission	Certificates assert a clean history, but omit an additional pre-cutoff final receipt; actual $Y_H = 0$. No hold before disclosure; hold afterward.	$\{1\}$	True	False	False

Evidence: [Research session 7, artifact 9]; claims: C6

9 Discriminating evaluation and limits of assurance

A discriminating evaluation must test the correctness of its observations and injected faults before attributing outcome differences to an agent or recovery mechanism. The proposed sequence therefore begins with independent oracle and injector validation, proceeds through deterministic boundary schedules and service-contract audits, and only then compares agent policies under controlled conditions. This ordering makes the analytical assumptions testable rather than treating them as established properties of an implementation.

The evaluation protocol should first freeze the workload, outcome rubric, unit of analysis, evaluators, and execution horizon. For the narrow external-effect workload, one logical debit request is evaluated by its authoritative effect count at the specified horizon H; exactly one applied debit satisfies the effect-count requirement, while acknowledgment receipt is recorded separately. Broader task success must additionally use prespecified end-state, safety, and deadline requirements.

Repeated executions should be organized into task clusters, with automated acceptance and independent outcome adjudication retained for each audited execution; crashes and timeouts remain attempted executions rather than disappearing from the denominator. [C31; [Research session 1]]

The supplied boundary schedules and conformance panels are analytical test specifications, not executed experiments. They establish neither implementation conformance nor observed agent failure rates, recovery benefits, or causal architectural improvements. Moreover, incomplete historical extracts and unavailable originals restrict the contribution to a scoped synthesis: omitted objections cannot be treated as resolved, and the supplied material does not constitute an exhaustive literature or benchmark assessment. [C20, C21; [Research session 7, artifact 6]]

The first implementation stage should independently test both the ledger oracle and the fault injector, especially their ability to distinguish a failure before commitment from acknowledgment loss after commitment. Final-state reads must be checked for freshness and noninterference, rather than assumed to reveal authoritative state without changing the task. The reference observer also needs a concrete capture-and-cutoff protocol covering durable effects, ingress receipts, isolation, and crash boundaries.

Independent retention is insufficient unless the observer actually captures every relevant event; agreement between the agent's evaluator and a reference judge cannot establish the rate of jointly accepted failures, so unsupported reference-error allowances must remain sensitivity parameters rather than reported calibration. [C30, C23; [Research session 1]; [Research session 7, artifact 6]]

Deterministic execution should then precede stochastic agent evaluation. The proposed seven path checks, six-case boundary experiment, and seventeen-case evidence panel address different obligations and should retain separate verdicts. Their execution should cover receipt/effect crash boundaries, stable versus regenerated operation keys, delayed effects, missing evidence, visible contradictions, and silently false certificates, while keeping routine evidence separate from held-out reference evidence.

Results should distinguish mapper conformance, service or controller contract failures, and failures of actual-outcome containment; otherwise a correctly implemented mapper operating on false evidence could be misdiagnosed as an implementation defect. [C30, C20; [Research session 1];

[Research session 5]; [Research session 7, artifact 6]]

A real-service audit is a separate applicability requirement, not a consequence of passing synthetic schedules. Before applying the atomic stable-key argument, the audit should establish effect/receipt atomicity, key namespace and lifetime, payload consistency, concurrent-call behavior, retention and expiry, response semantics, and downstream effects. Crash, failover, and restore checks should test whether the relevant durability guarantees survive the failures against which recovery is claimed. Where revocation or settlement progress is required, those properties likewise need direct validation rather than inference from the existence of an idempotency interface. [C85; [Research session 5]]

Only after these checks should the study conduct paired policy comparisons, fixing task definitions and weights, model and prompt versions, tool contracts, environment generation, deadlines, and resource budgets. Candidate comparisons include bounded retry, status reconciliation, safe escalation, restart, and reconciliation, but the analytical traces do not select a preferred policy. Shared fault schedules should be distinguished from mechanism-specific outage tests, and paired executions require isolation sufficient to prevent one policy's external effects from changing the other's task.

Sample sizes or precision targets and paired uncertainty estimates should be preregistered, with token, tool, latency, and monetary budgets reported separately because invocation count alone is not total cost. [C31; [Research session 1]; [Research session 5]]

The sampling design must preserve the distinction between independent task clusters and correlated repetitions of the same task. Repetition can characterize within-task variability, but it should not be counted as additional independent task sampling; cluster independence, population matching, and noninterference require evidence in the implemented evaluation. A finite-roster audit should separately validate roster completeness, randomized selection, evidence mapping, and bound calculation, including all-unknown evidence and outcome-dependent evidence loss.

Missing or inconsistent audits must remain unresolved under prespecified conservative treatment, while failed attempts, seeds, versions, tool traces, adjudications, and resource records should be retained rather than selected out of the comparison. [C31, C30; [Research session 1]; [Research session 5]; [Research session 7, artifact 6]]

Stress testing should explicitly vary correlated and persistent failures, delayed acknowledgments, transport duplication, concurrency, and deadline sensitivity rather than assume that an independent-attempt result transfers. Comparative outcomes should include regressions, severity, safe stopping, recovery quality, persistent-state corruption, delayed harm, and resource use alongside fixed-horizon success. Held-out task families and changed environments require separate evidence, as do multi-service workflows and settlement after H: exactly one effect for one logical operation does not establish end-to-end task correctness.

These measurements would determine whether a recovery mechanism improves the intended outcome under matched conditions, rather than merely increasing acknowledged completion. [C31, C85; [Research session 1]; [Research session 5]]

This evaluation sequence changes the assurance claim from an architectural expectation into a set of separable, falsifiable obligations. Conditional analysis identifies which observation and service premises would support a guarantee; deterministic tests can challenge an implementation's compliance with those premises; outcome-grounded, controlled comparisons are needed to establish an

improvement. Demonstrated deployment reliability would additionally require justified sampling and transport to the relevant tasks, environments, versions, and budgets—not merely successful execution of the boundary schedules.

Table 5: Proposed evaluation stages and the distinct assurance question each addresses. These are validation requirements, not completed tests or observed results.

Stage	**Required discriminating checks**	**Conclusion not warranted by this stage alone**
Oracle and injector validation	Distinguish precommit failure from postcommit acknowledgment loss; test fresh, nonmutating state reads and complete reference capture.	Evaluator agreement does not establish actual correctness or calibrate shared reference errors.
Deterministic boundary execution	Run the proposed path, recovery, and evidence panels; separately classify mapper, contract, and containment failures.	Passing finite cases does not prove universal coverage or a measured agent-level benefit.
Real-service contract audit	Check atomicity, key scope, payload binding, concurrency, retention, response semantics, durability, and downstream effects.	Synthetic conformance does not establish that an external service satisfies the recovery premises.
Controlled paired comparison	Freeze workload, versions, environment, rubric, horizon, and budgets; retain failures and unknowns; report paired uncertainty and broader harms.	A fixed-protocol comparison does not automatically transfer to changed tasks, environments, versions, or budgets.

Evidence: [Research session 1], [Research session 5], [Research session 7, artifact 6]; claims: C20, C23, C30, C31, C85

10 Limitations

- The analytical guarantees depend on unvalidated reference-error, evidence-soundness, atomicity, durability, identity, retention, and enforcement-scope premises.
- No implementation, benchmark audit, controlled agent comparison, or empirical cost analysis was completed.
- The irreversible-effect workload does not establish general reliability across open-ended agent tasks.
- The evidence selection is bounded and includes incomplete historical arguments.

- Saved research session a1a9ab51-f8d1-4b60-ba10-baad7585a34f ended with an incomplete review; call 8 (investigator) was not successfully completed. Earlier findings remain available, but this process limitation prevents claiming that the planned adversarial review was fully performed.
- Saved research session ebc7ac05-823c-4a02-9590-1e9f9726e807 ended with an incomplete review; a research review was not successfully completed. Earlier findings remain available, but this process limitation prevents claiming that the planned adversarial review was fully performed.
- Saved research session e3f02cce-aed5-42f7-aa01-a5cad3c99c18 ended with an incomplete review; call 19 (skeptic) was not successfully completed. Earlier findings remain available, but this process limitation prevents claiming that the planned adversarial review was fully performed.

11 Open obligations

- Recover complete proof notes and unresolved original objections before claiming historical compatibility or validated proofs.
- Validate directional reference-error allowances against justified external outcome evidence.
- Implement and independently test the authoritative outcome oracle and fault injector.
- Execute the crash, delayed-request, expiry, concurrency, cancellation, and false-certificate controls.
- Audit real-service atomicity, key scope, payload checks, retention, downstream effects, and acknowledgment semantics.
- Conduct controlled paired comparisons and report correctness, uncertainty, latency, resource consumption, and monetary cost without discarding unresolved executions.

12 Conclusions

Reliability evaluation must distinguish actual outcomes, reported completion, and the evidence used to certify each. Conditional statistical guarantees can accommodate correlated repetitions and imperfect judges, but their remaining truth-error assumptions require external validation.

Safe recovery depends on enforceable service contracts at the actual effect boundary. Stable keys, checkpoints, negative reads, and client stopping are not interchangeable with atomic suppression or durable execution-time cancellation.

The supported contribution is an analytical framework and proposed validation program. Demonstrated architectural benefit and deployment reliability remain open empirical questions.

A Claim and source notes

These notes identify the saved evidence for claims used in the article. The complete claim register and reasoning are retained in the paper record.

C15 Under the stated concurrent finite-retention contract, at-most-once safety survives overlapping workers and receipt garbage collection when current eligibility, receipt inspection, and effect-plus-receipt commitment share one per-operation ordering. For authorization interval $[a, D)$ with $D - a \leq T$, a first commitment at $c \geq a$ has suppression lasting through the remaining eligible interval because $c + T \geq D$. *Status:* unresolved. *Evidence:* [Research session 6, artifact 7]

C16 A negative linearizable status read does not make replacement safe when an outstanding original request can still execute. The legal ordering $\text{status}(r) < \text{commit}(r') < \text{commit}(r)$ produces two effects even though the negative read was accurate when performed. *Status:* established. *Evidence:* [Research session 8, artifact 4], [Research session 7, artifact 2]

C17 Under the specified delayed-or-discarded execution model, wrapper-local records, epochs, clocks, and absence replies cannot provide both sound conclusive cancellation and eventual authorization of replacement if the downstream effect remains outside the wrapper's enforcement boundary. *Status:* unresolved. *Evidence:* [Research session 8, artifact 5]

C18 For an initially unperformed operation, a durable cancellation acknowledgment certifies no prior or future effect if cancellation and actual effect commitment are competing crash-atomic transitions from a pending state, every effect-producing path obeys that gate, and suppression covers every later execution opportunity. *Status:* unresolved. *Evidence:* [Research session 6, artifact 6]

C19 An atomic durable closure gate prevents additional covered effects after closure commits, even when historical effects already exist. It does not remove prior duplication, establish compensation, prove notification receipt, or contain already accepted downstream work outside the enforced boundary. *Status:* unresolved. *Evidence:* [Research session 7, artifact 7]

C20 The saved boundary schedules and fault-injection panels are proposed analytical tests, not executed experiments. The supplied evidence establishes no implemented harness conformance, observed agent failure rate, measured recovery benefit, or causal architecture improvement. *Status:* limitation. *Evidence:* [Research session 1], [Research session 5], [Research session 7, artifact 6], [Research session 6, artifact 5]

C21 The available evidence is a bounded selection containing incomplete historical extracts and unresolved continuity obligations. It supports a scoped analytical synthesis, not an exhaustive literature assessment or a claim that all earlier objections have been resolved. *Status:* limitation. *Evidence:* [Research session 2], [Research session 3], [Research session 8], [Research session 7, artifact 9]

C22 Subtracting two lower confidence bounds on individual reliability does not establish a lower bound on improvement. A valid improvement argument must also control how much baseline correctness can exceed its observed or certified surrogate. *Status:* established. *Evidence:* [Research session 5, artifact 1], [Research session 5, artifact 3]

C23 Agreement between the agent's evaluator and a reference judge does not establish a useful bound on jointly accepted failures. Without justified external outcome evidence, reference-error allowances must be reported as sensitivity parameters rather than measured calibration. *Status:* limitation. *Evidence:* [Research session 4, artifact 1], [Research session 2], [Research session 3]

C13 Stable operation identity and atomic effect-plus-receipt commitment do not ensure at-most-once effects after suppression expires. In the saved constructed schedule, an effect commits at time 1, a retention

duration of 10 expires at time 11, and an outstanding same-key request executes at time 12, creating another effect. *Status:* established. *Evidence:* [Research session 6, artifact 4], [Research session 6, artifact 5]

C14 Two stated contracts suffice to restore at-most-once safety with finite result retention: suppression that remains effective through every possible effect-capable execution, or durable rejection state that survives result expiry and blocks later execution. These are safety conditions, not guarantees that an effect will occur. *Status:* unresolved. *Evidence:* [Research session 6, artifact 4], [Research session 6, artifact 5]

C12 In the refined destructive-dequeue model, receiving a request removes its delivery envelope from the durable queue and places execution only in volatile worker state. A crash loses that consumed worker without reconstructing its delivery, while envelopes still queued survive; commitment and acknowledgment enqueueing are separate transitions. *Status:* established. *Evidence:* [Research session 4, artifact 4]

C11 Effect-first and marker-first split-write handlers fail in different ways. Effect-first commitment followed by a crash before recording completion permits duplication on retry; marker-first suppression can retain at-most-once safety while omitting the effect and potentially returning a cached success. *Status:* established. *Evidence:* [Research session 7, artifact 3], [Research session 7, artifact 5], [Research session 6, artifact 3]

C10 For an initially unperformed logical operation, a stable key and serialized, durable, atomic effect-and-completion-record insertion preserve at-most-one covered effect through retries and crashes. Exactly one effect additionally requires a valid request to commit; eventual verified client completion also requires suitable response delivery and durable client completion. *Status:* unresolved. *Evidence:* [Research session 4, artifact 2], [Research session 4, artifact 3], [Research session 6, artifact 2]

C85 A real-service application of the atomic stable-key argument requires checking effect/receipt atomicity, key scope, payload consistency, concurrency, retention, response semantics, and downstream effects. *Status:* unresolved. *Evidence:* [Research session 5]

C31 The intended experiment compares outcomes under fixed task, model, prompt, tool, environment, budget, rubric, and horizon definitions. *Status:* unresolved. *Evidence:* [Research session 1, artifact 1], [Research session 8, artifact 1]

C30 The practical test program must first establish oracle and fault-injector correctness before using their outputs to compare agents. *Status:* unresolved. *Evidence:* [Research session 1]

C9 Relaxing guaranteed retry commitment invalidates the original retry-probability bound. *Status:* established. *Evidence:* [Research session 5]

C1 Reliability concerns actual rubric satisfaction under a fixed execution protocol, not completion reports alone. *Status:* established. *Evidence:* [Research session 1, artifact 1]

C2 If jointly accepted failures have probability at most η, then $p \geq \max(0, \Pr(O = 1) - \Pr(O = 1, R = 0) - \eta)$. *Status:* unresolved. *Evidence:* [Research session 4, artifact 1]

C3 Repeated runs need not be independent; task-cluster sampling determines the stated concentration guarantee. *Status:* unresolved. *Evidence:* [Research session 1, artifact 1]

C4 Paired improvement bounds require directional truth-error corrections for candidate false successes and baseline missed successes. *Status:* unresolved. *Evidence:* [Research session 5, artifact 3]

C5 Random auditing can conditionally certify a frozen paired roster without independent execution outcomes. *Status:* unresolved. *Evidence:* [Research session 6, artifact 1]

C6 Mapper conformance, evidence validity, and actual-outcome containment are distinct properties. *Status:* unresolved. *Evidence:* [Research session 7, artifact 9]

C7 A success report at tick 15 followed by the sole correct commit at tick 18 can pass a horizon-20 end-state rubric while being false when emitted. *Status:* established. *Evidence:* [Research session 7, artifact 8]

C8 Under forced stop-or-retry recovery with guaranteed retry commitment, indistinguishable committed and uncommitted histories imply worst-case error at least 1/2. *Status:* unresolved. *Evidence:* [Research session 1, artifact 2]

B Editorial writing sources

These sources informed manuscript organization and writing methods, not the scientific evidence or findings.

- PLOS ONE — Manuscript and Reporting Guidelines (general): `https://journals.plos.org/plosone/s/submission-guidelines`
- EASE — Guidelines for Authors of Scientific Articles (full text) (general): `https://pmc.ncbi.nlm.nih.gov/articles/PMC4130677/`